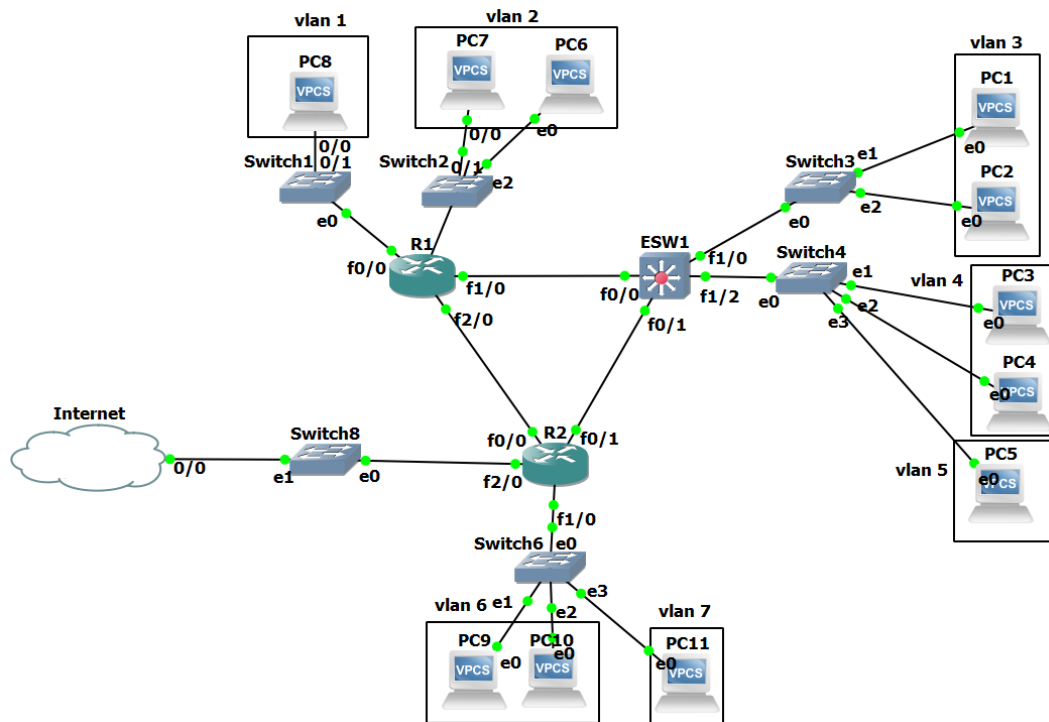




Assume that you are the manager of the above network, which has Internet connectivity either directly (public IPv4), through NAT/PAT on router R2, or via global IPv6. Assume that routing and gateways are correctly configured. Regarding its public IPv4 addressing, the network received from its ISP a portion of a public Class C IPv4 network range: **202.13.44.0/26**. These addresses are distributed according to needs:

- **NAT/PAT:** 3 addresses
- **VLAN 3:** 2 addresses
- **VLAN 4:** 12 addresses
- **VLAN 5:** 5 addresses
- **VLAN 6:** 1 address
- **VLAN 7:** 8 addresses

The network also uses private IPv4 addresses internally. The network used is **192.168.16.0/21**. The number of addresses required at each location is:

- **VLAN 1:** 20 addresses
- **VLAN 2:** 825 addresses
- **VLAN 3:** 25 addresses
- **VLAN 4:** 180 addresses
- **VLAN 5:** 5 addresses
- The interconnections between routers and the ESW also use private IPv4 addressing from the same network. There is a DHCP server configured to assign private addresses to the networks.

Regarding IPv6, the network received from its ISP the following IPv6 network: **2002:1234:5678:0000::/56**, which is distributed as follows:

- Each router (including the ESW) manages a different **/60** subnetwork, derived from the **/56**.
- Each VLAN must have a **/64** network, taken from the subnetwork assigned to the router/ESW that provides it with connectivity.
- The interconnections between routers use global IPv6 **/126**, taken from a **/60** subnetwork (different from the **/60** subnetwork associated with each router).

1. (5 points) Considering the previous network:

a) Define private IPv4 subnets (identifier and mask) for the VLANs, including the point-to-point links. Justify your answer (with schematics). (2 points)

R: vlan2: 192.168.16.0/22; vlan4: 192.168.20.0/24; vlan1: 192.168.21.0/27; vlan3: 192.168.21.32/27; vlan5: 192.168.21.64/29; p2p: 192.168.21.72/30, 192.168.21.76/30, 192.168.21.80/30.

b) Define public IPv4 subnets (identifier and mask) for the VLANs, including the NAT/PAT addresses. Justify your answer (with schematics). (2 points)

R: vlan4: 202.13.44.0/28; vlan7: 202.13.44.16/28; vlan3: 202.13.44.32/29; vlan5: 202.13.44.40/29; vlan7: 202.13.44.48/30; NAT/PAT: 202.13.44.52/30.

c) Define IPv6 subnets (identifier and mask) for the VLANs, including the point-to-point links. Justify your answer (with schematics). (1 point)

R: vlan1 and vlan2: 2002:1234:5678:0000::/64 e 2002:1234:5678:0001::/64;
Vlan3, vlan4, vlan5: 2002:1234:5678:0010::/64, 2002:1234:5678:0011::/64, 2002:1234:5678:0012::/64;
Vlan6 and vlan7: 2002:1234:5678:0020::/64, 2002:1234:5678:0021::/64;
P2p: 2002:1234:5678:0030::/126, 2002:1234:5678:0030::4/126, 2002:1234:5678:0030::8/126.

2. (3.5 points) Considering the previous network, and initially with empty ARP and switching tables:

a) Is it possible to have communication between PC3 and PC5 using only Switch4? And between PC3 and PC4? Justify. (1 point)

R: PC3 and PC5: no (switch does not route different vlans); PC3 and PC4: yes.

b) Considering the situation in which PC3 and PC5 (case A) and PC3 and PC4 (case B) are able to communicate, are there packets exchanged between Switch4 and ESW? If so, which ones in each case? Justify. (1 point)

R: PC3 and PC5: all packets; PC3 and PC4: only the first ARP Request.

c) What is the Switch4 switching table after these two communications? Justify. (1.5 points)

R: MAC PC3, e1; MAC PC4, e2; MAC PC5, e3; MAC F1/2 ESW, e0.

3. (6.5 points) In a communication between PC3 and PC7, consider empty switching and ARP tables.

a) Which packets are captured on PC5 and PC6? Justify. (1.5 points)

R: PC5: nothing; PC6: First ARP Request from R1 to Broadcast, searching for PC7.

b) If the user of PC3 makes a mistake and tries to communicate with a non-existent address in VLAN 2, what happens (which packets, if any) in VLAN 4 and in VLAN 2? Justify. (1.5 points)

R: vlan4: ARP Req+Rep; ICMP Request; vlan2: ARP Request from R1 to Broadcast, searching for the non-existent address; no answer and there will be timeout.

c) Considering static routing, what is the private routing table on the ESW for the internal networks and the Internet? Justify. (1.5 points)

R: vlan 3, 4 and 5 directly connected; vlan1 and 2 through f1/0 R1; vlan6, 7 and the Internet through f0/1 R2.

d) If the user of PC3 makes a mistake and tries to communicate with IP 200.13.44.80, what will happen? Justify. (1 point)

R: It will go through the default route in ESW to the Internet; Host unreachable will be received.

e) Without access to the router's NAT/PAT table, and using only Wireshark, if PC3 communicates with a PC on the Internet, how can you verify that NAT/PAT is working correctly (which link you should monitor the packets on, and what you expect to see in those packets)? Justify. (1 point)

R: Connection to the cloud, sender address of the outgoing packets with public NAT/PAT address.

4. (1.5 points) When two terminals connect to a wireless network through an access point, what is their reaction when they detect collisions? When can they communicate again? Justify.

R: Stop transmitting and wait a random time before transmitting again, which increases exponentially every trial.

5. (3.5 points) Consider that a given PC has three services: voice at 64 kb/s, video at 1 Mb/s, and the download of a 500 MB file.

a) In TCP (for the file download), how is it determined that the time to receive an acknowledgment for a packet delivery has expired? Justify. (1 point)

R: It depends on the round trip time (its mean and variance), to make sure it waits more time for the packet than the round trip time.

b) If at the beginning of the connection only the first and last services are active, and the video service starts halfway through, with limited link capacity, how will the network react and what impact will this have on the different services? Justify. (1.5 points)

R: Losses; voice will not be impacted; download will decrease the rate due to TCP congestion window, and accommodate the video.

c) If you also integrate a web page access service using HTTP 1.1, is it possible to have a connection with pipelining and without persistence? Justify. (1 point)

R: No; pipeline requires that more than 1 object can be transmitted in each TCP connection.